# Vehicle Design Aerodynamics

## SolidWorks Flow Simulation Sports car model Report

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## 1. Sketch views of the concept

The objective of the sports car is to travel as fast as possible, so the characteristics of the design had to be streamlined and smooth. The car hood and nose are quite rounded to minimise air drag. Additionally, the windscreen, roof and boot were all streamlined and designed smaller than commercial cars to decrease air resistance by providing an aerodynamic surface for the air to glide over.

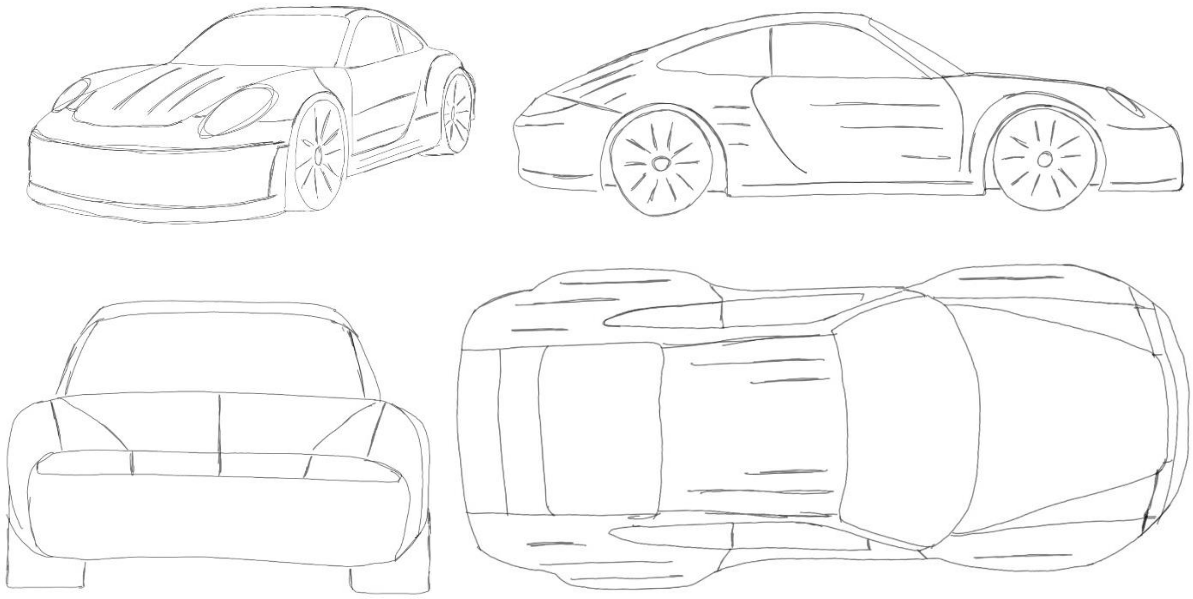

*Figure 1_ Concept sketching*

## 2. CAD model

Cad model: [AranchaRamirez_carmodel.STEP]

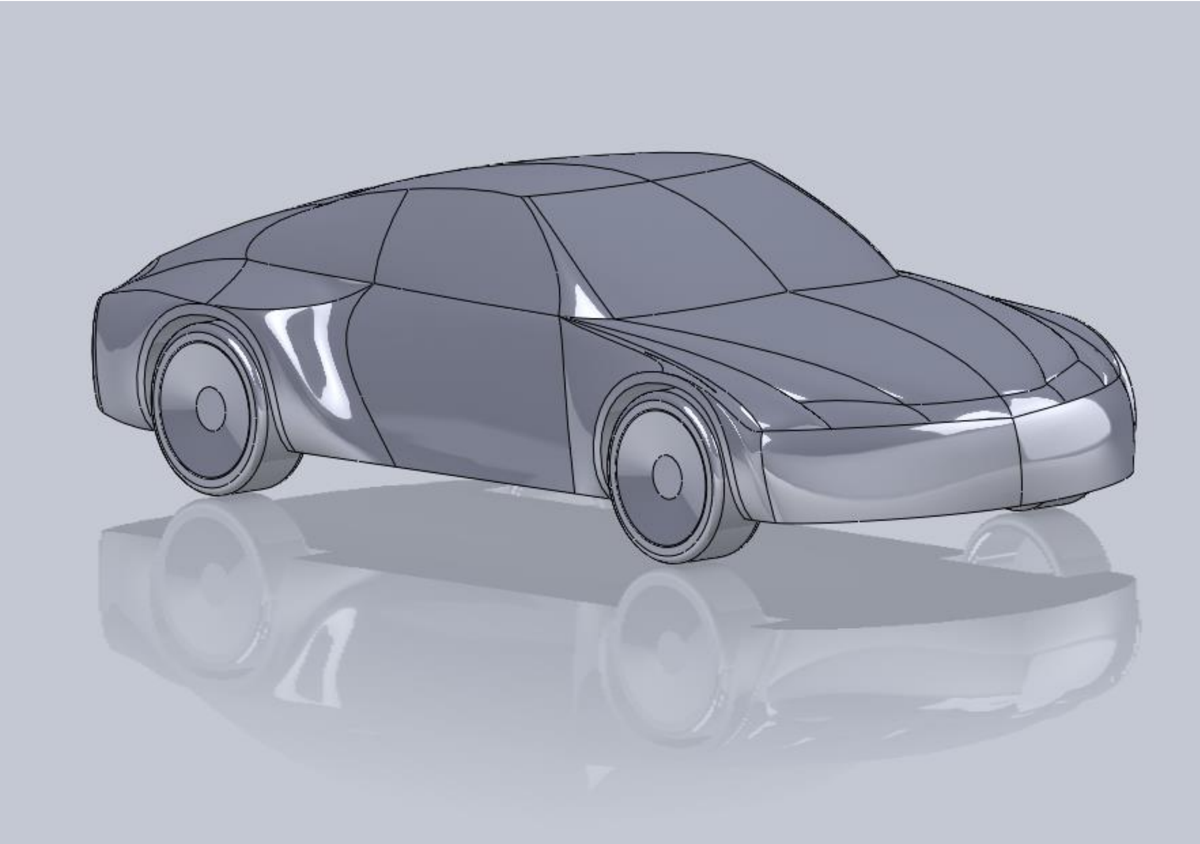

*Figure 2_CAD model*

## 3. Drag analysis and preliminary hand calculations

Using the provided formulas, an estimate for the aerodynamic drag of my concept car was calculated.

Considering the following body styles and their values (from Figure 4.15 and Table 4.5 on Drag notes):

| Component | Shape | N |
|---|---|---|
| Front end (top) | A-1 | 1 |
| Windshield | B-2 | 2 |
| Roof | C-2 | 2 |
| Lower rear end | D-2 | 2 |
| Front end (side) | E-5 | 3 |
| Windshield peak | F-1 | 1 |
| Rear roof/trunk | G-1 | 1 |
| Cowl/fender cross section at windshield | H-3 | 3 |

Considering the resulting drag coefficient expressible as:

$$C_D = 0.16 + 0.0095 \sum_{i=A}^{H} N_i$$

Where $N_i$ is the numerical values associated with the 8 shapes identified car components,

$$C_D \ = \ 0.16 + 0.0095 * \ (1 + 2 + 2 + 2 + 3 + 1 + 1 + 3)$$

$$C_D = \ 0.3025$$

Re-arranging drag coefficient expression:

$$C_D = \frac{Drag}{0.5\mathrm{pU}_{\infty}^{2}\mathrm{A}}$$

$$Drag = C_D * (0.5pU_{\infty}^{2}A)$$

Were,

$$density\ of\ air, p = 1.225kg/m^3$$

$$U_{\infty} = 30m/s$$

$$A = 1.76\ m^2$$

$$Drag = 0.3025 * (0.5 * 1.225 * (30)^2 * 1.76)$$

$$Drag \ = \ 293.4855N \ = 0.293kN$$

# 4. CFD model

## 4.1 CFD setups and initial conditions

**Analysis of Mesh**

Total Cell count: 505,417 1,184,649

Fluid Cells: 505,417 1,184,649

Fluid Cells contacting solids: 18,613

Trimmed Cells: 0

**Additional Physical Calculations option**

Heat Transfer Analysis: Heat conduction in solids: Off

Flow Type: Laminar and Turbulent

Time-Dependent Analysis: Off

Gravity: On

Radiation:

Humidity: Off

Default Roughness: 0 micrometre

**Material Settings**

Fluids: Air

**Initial Conditions**

*Table 1_Ambient Conditions*

| | |
|---|---|
| Thermodynamic parameter | Static Pressure: 101325.00 Pa<br>Temperature: 293.20K |
| Velocity parameters | Velocity vector<br>Velocity in X direction: 0 m/s<br>Velocity in Y direction: 0 m/s<br>Velocity in Z direction: -30.000 m/s |
| Turbulence parameters | Turbulence intensity and length Intensity: 0.10%<br>Length: 6.689e-04 m |

**Engineering Goals**

Global Goals:

*Table 2_Velocity*

| | |
|---|---|
| Type | Global Goal |
| Goal type | Velocity (Z) |
| Calculate | Average value |
| Coordinate system | Global Coordinate System |
| Use in convergence | On |

*Table 3_Lift force*

| | |
|---|---|
| Type | Global Goal |
| Goal type | Force (Y) |
| Coordinate system | Global Coordinate System |
| Use in convergence | On |

*Table 4_Drag force*

| | |
|---|---|
| Type | Global Goal |
| Goal type | Force (Z) |
| Coordinate system | Global Coordinate System |
| Use in convergence | On |

*Table 5_Drag coefficient equation*

| | |
|---|---|
| Type | Equation Goal |
| Formula | (2*Drag Force) /(1.225*0.0061(Velocity^2)) *(-1) |
| Dimensionality | No units |
| Use in convergence | On |

**Analysis Time**

Calculation Time: 1035

Number of Iterations: 271

**Results**

*Table 6_Outcomes*

| Name | Unit | Value | Progress | Criteria | Delta | Use in convergence |
|---|---|---|---|---|---|---|
| Velocity | $m/s$ | -30.000 | 100 | 0.0 | | On |
| Lift Force | $N$ | 459.967 | 100 | 7.990 | 7.930 | On |
| Drag Force | $N$ | -359.232 | 100 | 35.308 | 1.758 | On |
| CD | $n/a$ | 0.340 | 100 | 0.002 | 0.033 | On |
| Frontal area | $m^2$ | 1.76 | | | | |

## 4.2 CFD analysis and plots

The flat front end of the car results in the highest-pressure region.

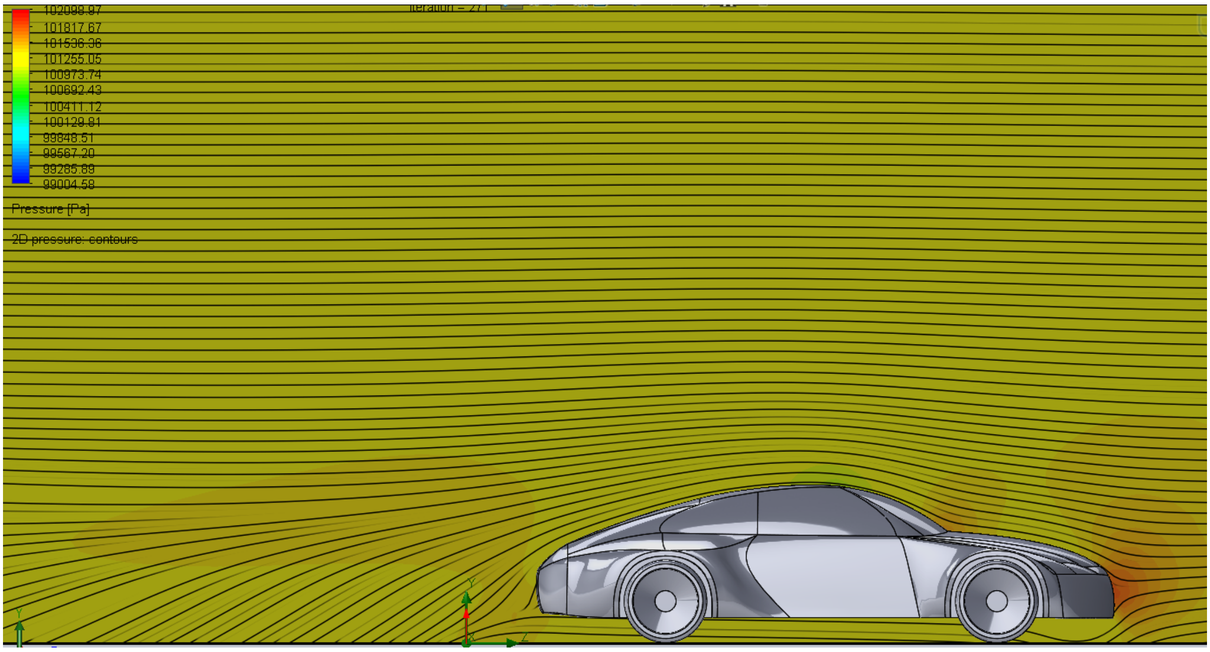


*Figure 3_ pressure plot 2D*

The recirculation zone, represented in blue behind the vehicle, is relatively small which reduces drag.

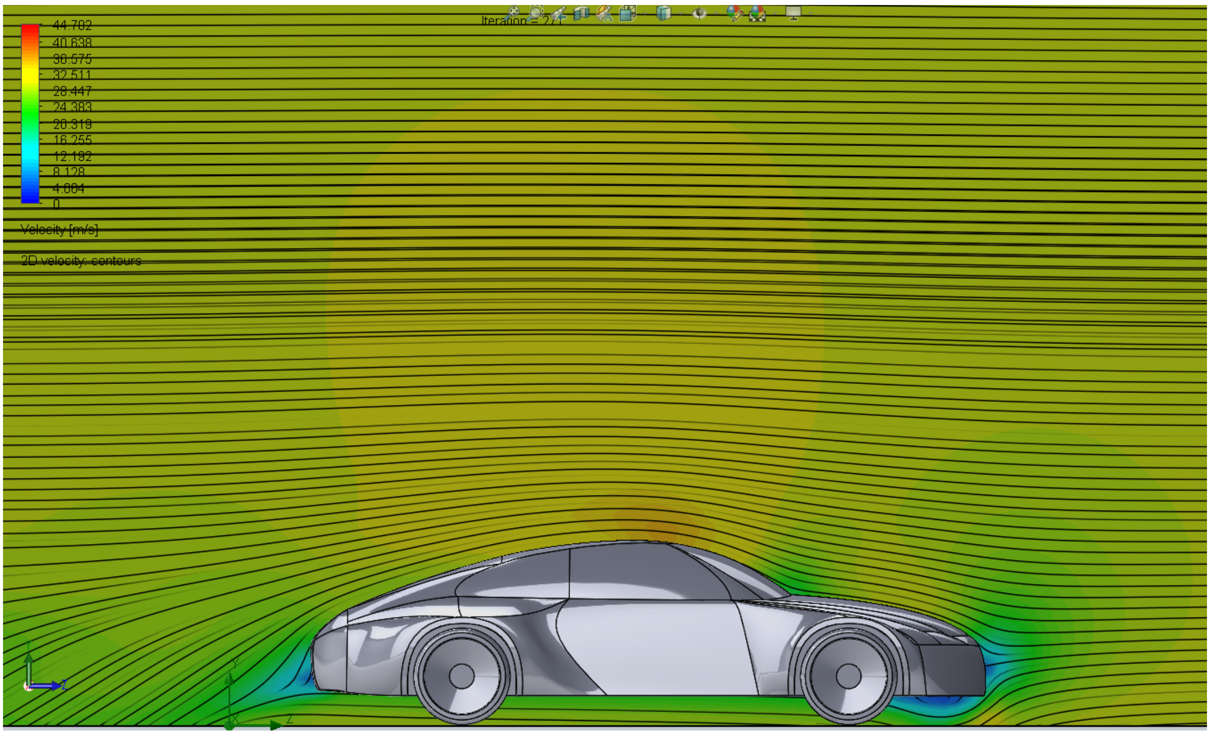


*Figure 4_2D velocity plot*

Additionally, some high pressure can also be observed at the bottom of the windscreen where all the air coming from the initial contact with the nose of the car crashes again into another surface. However, it experiences a lower pressure as the surface is smoother and more inclined instead of fully flat.

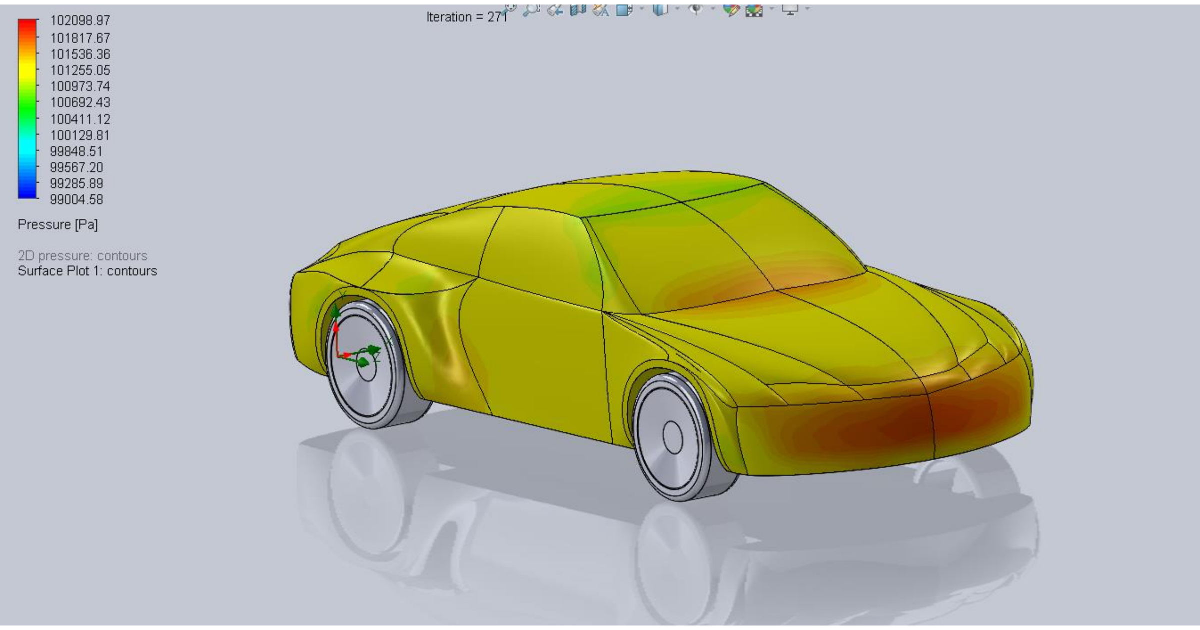


*Figure 5_ 3D pressure distribution (front view)*

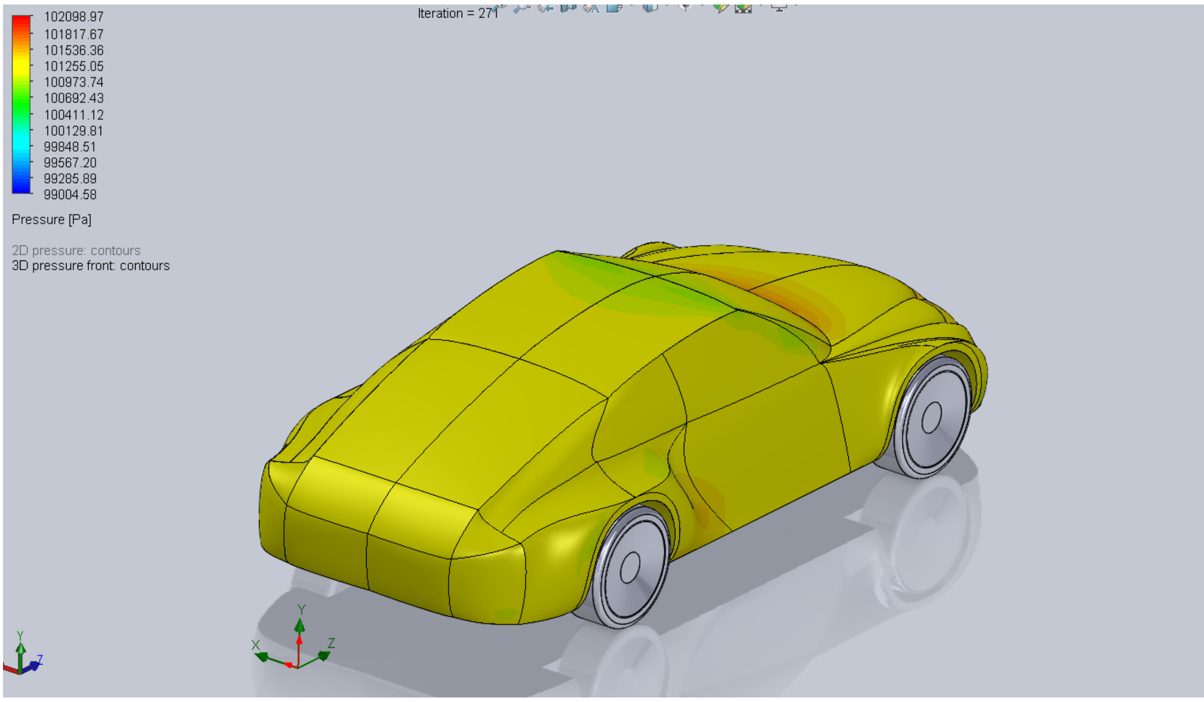


*Figure 6 _ 3D pressure distribution (back view)*

From the plots below, there is a higher air resistance at the nose and windscreen of the car, when the air first contacts the vehicle. However, there is less turbulence at the end, which is advantageous, as the longer the flow remains linked to the body the more drag is avoided.

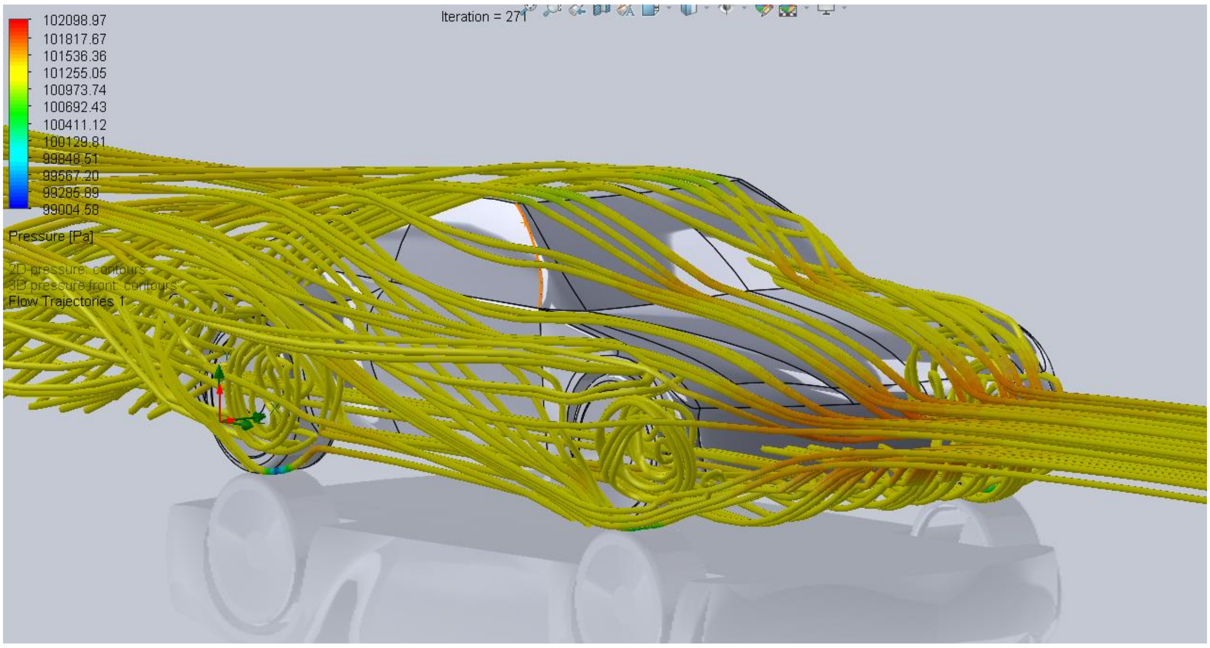


*Figure 7 _ Flow trajectories (front view)*

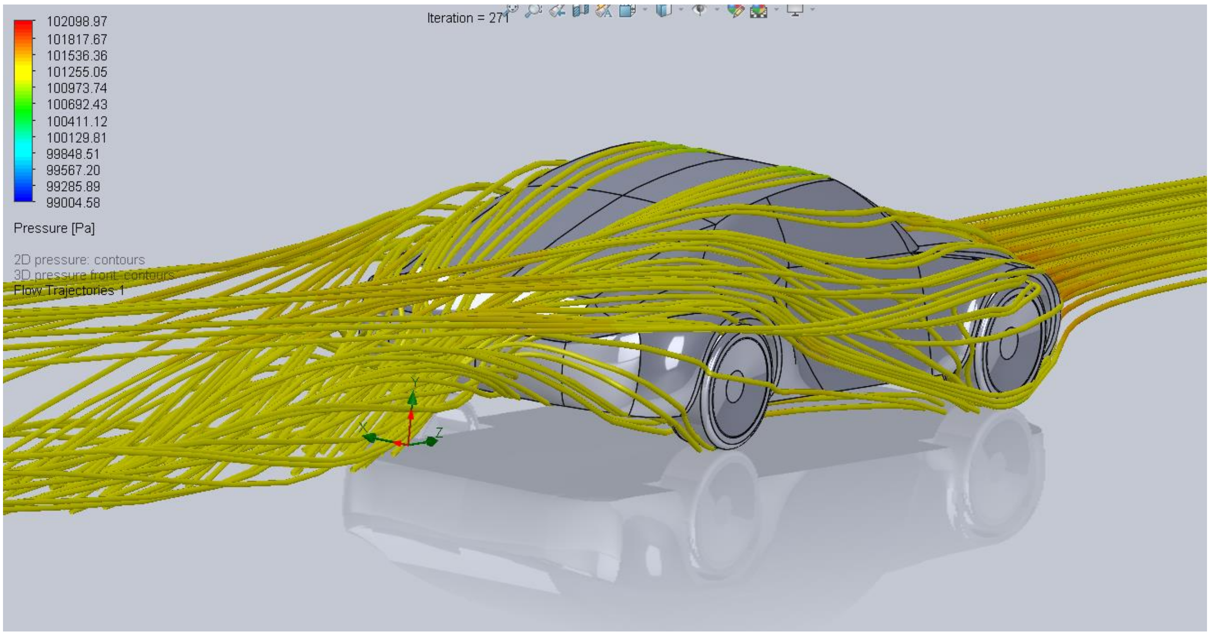


*Figure 8 _ Flow trajectories (back view)*

# 5. Discussion and Conclusion

## 5.1 CFD Comparison

**Deficit:**

The front end of my design has higher pressure due to the flatness of the surface. This increases wind resistance and channels large amounts of air underneath the vehicle, obstructing the air flow.

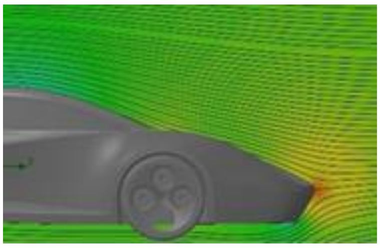

*Figure 9_ 2D pressure plot of the sports car*

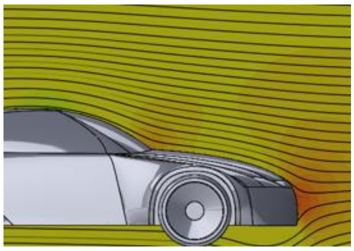

*Figure 10_2D pressure plot of my design*

**Upgrade:**

The recirculation zone achieved with my design is quite lower compared to the one observed in the sports car design, which results in less turbulence.

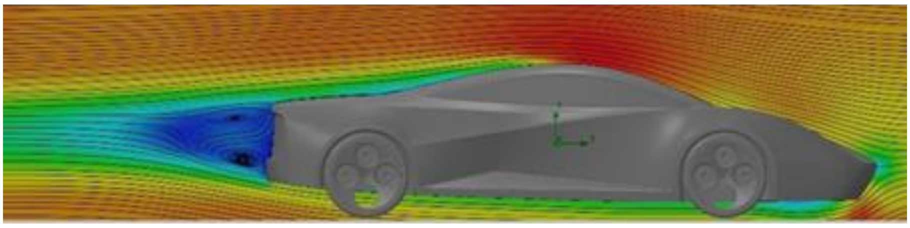

*Figure 11_2D velocity plot of the sports car*

Additionally, the wind tunnel test indicates increased turbulence at the rear of the vehicle, resulting in heightened drag and influencing the vehicle's drag coefficient. The wind tunnel test results indicate an increased drag coefficient by 0.1 compared to the one predicted by its CFD analysis.

*Figure 12_ Side view of sports car in the wind tunnel*

The percentage difference between the CFD drag coefficient for my design and the drag coefficient given by the wind tunnel test results:

$$Percentage\ differene\ =\ \frac{0.37 - 0.34}{0.37} x\ 100\ =\ 8.1\%$$

Overall, the drag coefficient of the sports car (0.37) is higher than the one achieved with my design (0.34). However, it should be noted that the CFD calculation is based on 'perfect' simulation calculated with a computer, while the wind tunnel test results are generally accepted as being more accurate, as they better portray the unpredictable inconsistencies that occur in the real world.

## 5.2 Conclusion

The most significant difference in the drag coefficient between CFD and wind tunnel tests are results from the front end of the vehicle. It is unavoidable to have a certain pressure at the front since it's the first point of contact with the air, but the air drag coefficient could be much lower if that surface was less flat and more aerodynamic to minimise air drag and better redirect air flow.

## 5.3 Future Improvements

The windscreen volume could be more rounded not only over the roof but also at the front to allow better air flow around the car.

Additionally, changing the front flat shape of the car's nose, to a hard-edge shape will separate the airflow upon impact, which will improve the stagnation point and reduce the drag force and better direct air flow.

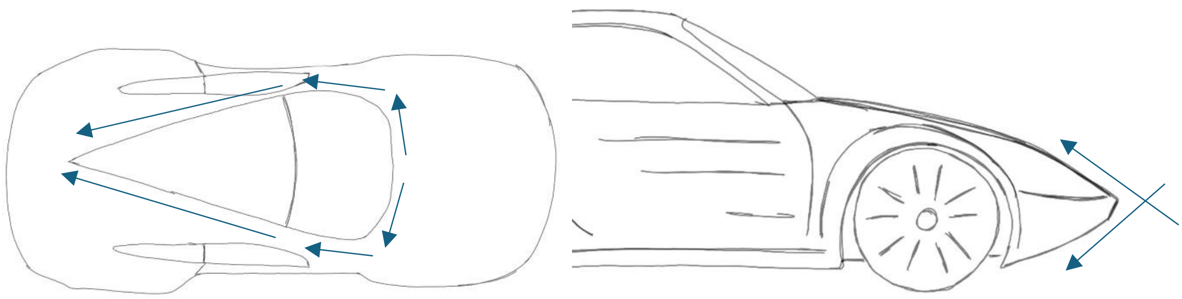

*Figure 13_Future car design airflow*

Additionally, body panels channelling the air could be added to the car to reduce wakes at the back, and front air dams could be added to the front to increase the downward force and decrease drag by decreasing the amount of air flowing under the car.